Q.26 Explain the importance of accuracy and repeatability for robot selection.

Q.27 Define a servo control system and explain how it uses feedback.

Q.28 How can adaptive control be used in collaborative robots.(cobots)?

Q.29 Explain how device controllers translate high-level commands into physical actions.

Q.30 State and explain two major industrial applications benefiting from advanced robot programming.

Q.31 Briefly explain why robot programming is essential in automation industries.

Q.32 Compare RAIL with VAN in the context of application and complexity.

Q.33 Explain the process of material transfer in robotics and give one practical example.

Q.34 How do painting robots reduce environmental hazards in factories.

Q.35 What types of sensors are commonly used in inspection robots and what do they achieve?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Discuss future challenges in robotics related to sensing, control, and artificial intelligence.

Q.37 Explain what is meant by manual teaching in robot programming, and describe how it is done. What are the advantages and limitations of manual teaching?

Q.38 Write the common benefits of using robots for material transfer, painting, packaging, inspection, and welding in industries.

(20)

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Roll No. ....

6th Sem / Mecatronics, Mech(CAD/CAM Design & Robotics)

Subject:- Robotics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Robots help to:

a) Increase worker injuries
b) Reduce risk exposure
c) Spend more resources
d) Make errors

Q.2 Hybrid control systems:

a) Use both open and closed loop
b) Only open-loop
c) Only closed -loop
d) None of the above

Q.3 Suction cups are often used to handle :

a) Metal parts b) Perforated materials
c) Nonporous surfaces d) Paper

Q.4 Which of these is NOT a common robot drive type?

a) Electric b) Hydraulic
c) Pneumatic d) Magnetic

Q.5 What is the primary purpose of robot control systems?

a) To physically build robots
b) To manage and coordinate robot movements and actions

c) To sell robots
d) To design robot body parts

Q.6 Limitations of adaptive control include:
a) High computational cost
b) No benefit in uncertain environments
c) Requires no sensors
d) Fixed parameters setting

Q.7 VAN language was developed for:
a) Painting robots only
b) Unimation robots
c) Space rovers
d) Autonomous drones

Q.8 RPL is designed for:
a) Non-programmers for plant automation
b) AI-Based high-end labs only
c) Space navigation
d) Satellite control

Q.9 One key benefit of robotic machine loading / unloading is:
a) Inconsistent cycle times
b) Increased production rate and safety
c) Higher manual labor cost
d) Less accuracy

Q.10 The main benefits of packaging robots is :
a) Random product placement
b) High-speed and accurate packing
c) Increased manual labor
d) Wastage of wrapping material

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Manipulator is a part of a robot used for_____ objects.

Q.12 RTD sensors are used for measuring________

Q.13 Electric drives are the most popular for_______ robots.

Q.14 SCARA robots are mainly used for ______ tasks.

Q.15 Resolved motion control is particularly useful for multi-axis robots. (True/False)

Q.16 VAN and RAIL are high-level general-purpose programming language. (True/False)

Q.17 Explain lead through programming.

Q.18 Define robot programming in your own words.

Q.19 Name two common types of robotic welding.

Q.20 Give one example of a material transfer robot.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain how robots increase quality and consistency in production process.

Q.22 What is linear motion in robotics, and where is it commonly used?

Q.23 Describe the classification of robotic system based on their physical structure?

Q.24 What are the advantages and limitations of mechanical grippers?

Q.25 What is a thermocouple. Explain why thermocouples are used for temperature measurement in robotic systems.